2025 IEEE International Conference on Intelligent Rail Transportation (ICIRT)
October 11-12, 2025. Beijing, China

An Optimization Method of Heavy Haul Train Driving Strategy Based on Dueling DQN

Qida Si[1], Lu Jiang[2], Qi Wang[2], Youbin Su[3],Zhiming Xiao[3],Yuxuan Han[4] and Ming Chai*[1]

***Abstract*—The operation of heavy haul trains on long and steep downhill is very challenging, it is difficult for drivers to accurately grasp the timing of the application of circulating air brakes. To address this problem, this paper proposes an optimization method for heavy haul train driving strategy based on the Dueling DQN algorithm. On the premise of ensuring safety, the optimization goal is to improve the efficiency of train operation and reduce the total time of air braking. On the premise of considering the time-delay characteristics of train air braking force, a multi-particle model of a heavy haul train is established, and then the optimization problem of the train driving strategy is transformed into a Markov decision process (MDP). According to the characteristics of heavy haul train operation, the key elements of Dueling DQN are designed, and the reward function is set based on the optimization objective. The simulation experiment is carried out in the scene of a long and steep downhill. At the initial speed of 50 km/h, the average speed of the whole train can reach 65.95 km/h. Finally, the optimization effect of the DQN algorithm is compared to verify the effectiveness of the proposed algorithm.**

I. Introduction

Heavy haul railway plays a vital role in China 's transportation with its strong carrying capacity and extremely high transportation capacity. With the rapid growth of China 's freight demand, the heavy haul railway is also developing in the direction of long marshalling and large weight. At present, the driving control of heavy haul railway in China mainly depends on the driver 's manual control. However, the complex line environment of heavy haul railway makes it much more difficult for heavy haul trains to drive than ordinary trains, and it is easy to cause safety accidents by relying solely on manual experience. In addition, in the long and steep downhill section, the driver needs to perform circulating air braking to ensure that the train speed does not exceed the limit ; due to the constraints such as air filling time, the circulating air brake has high requirements for the driver. If the operation is improper, it can easily cause the train to stop, which will seriously affect the transportation capacity and economic benefits. Therefore, it is necessary to analyze and study the driving strategy optimization problem of heavy haul trains according to the characteristics of heavy haul trains, so as to improve the train operation speed, reduce air braking time, and thereby improve the efficiency of heavy-haul rail transport.

As an important branch of machine learning, reinforcement learning (RL) plays an important role in intelligent decision-making research. During the training process, the agent interacts with the environment continuously, and gradually learns the optimal action strategy in continuous trial and error. Reinforcement learning can effectively adapt to the optimization needs of discrete and continuous action space. In recent years, it has been widely used in video games [1], automatic driving [2], robots [3] and other fields. When studying the optimization problem of heavy haul train driving strategy, traditional optimization algorithms such as mathematical programming, dynamic programming, and heuristic algorithms are difficult to express complex air braking constraints, and the optimal driving strategies are mostly non-long downgrades. In the problem of heavy haul train driving strategy, the train action is discrete and the state is continuous. It is suitable for reinforcement learning to optimize, and can save the complex modeling process. Compared with the traditional optimization algorithm, it is more flexible.

In this paper, aiming at the operation problem of long and heavy haul trains in long and steep downhill sections, under the premise of ensuring safety, the corresponding dynamic model is established to improve the train operation efficiency and reduce the total air braking time. The Dueling DQN algorithm is used to optimize the operation strategy and operation curve of the train in the long and steep downhill section. The main contributions are as follows :

- Since the time-delay characteristics of train air braking force are not considered in [4,5,8], a multi-particle model of a train considering the time-delay characteristics of air braking force is established to make the simulation more in line with the actual process.
- Dueling DQN algorithm can better evaluate the value of the state and accelerate the training speed. We use this

This work was partially supported by the Fundamental Research Funds for the Central Universities (22JBQY001, 2025JBXT011); National Natural Science Foundation of China (372309, 52172322, U22A2046, 52272329, U2468208, U2468206, U2469211, 52372310, U246920087), Beijing Natural Science Foundation, China (231001, L251033, L251056); State Key Laboratory of Advanced Rail Autonomous Operation (RAO2023ZZ001);Technology Innovation Project of CHN Energy(No.GJNY-22-82);The subject of the science and technology research development plan of China Railway Corporation (N2023G058); Ministry of Industry and Information Technology of PRC (JTGXB-001), Beijing Laboratory of Urban Rail Transit.(Corresponding author: Ming Chai)*

[1]Qida Si and Ming Chai are with National Engineering Research Center of Rail Transportation Operation and Control System, Beijing Jiaotong University, 100044, Beijing, China. Corresponding author: Ming Chai Email: chaiming@bjtu.edu.cn.

[2]Lu Jiang and Qi Wang are with China Academy of Railway Sciences Limited Corporation, Beijing, 100081, China; 20120252@bjtu.edu.cn.

[3]Youbin Su and Zhiming Xiao are with CHN Energy Shuohuang Railway Development Co., Ltd, 062350, Hebei, China; 2507804943@qq.com.

[4]Yuxuan Han is with Beijing No.159 High School,2536563271@qq.com.

2025 IEEE International Conference on Intelligent Rail Transportation (ICIRT) | 979-8-3315-9751-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICIRT66379.2025.11216735

Authorized licensed use limited to: University of Pittsburgh. Downloaded on July 07,2026 at 23:51:09 UTC from IEEE Xplore. Restrictions apply.

algorithm to optimize train operation and design the key elements of the algorithm.

- The simulation experiment is carried out in a slope environment with a slope of -10.2 ‰, and the optimization effect of DQN is compared to verify the effectiveness of the proposed algorithm.

The structure of this paper is as follows : Section II introduces the related research on curve planning and train operation optimization using reinforcement learning algorithm. In Section III, a driving strategy optimization model of heavy haul train based on Dueling DQN is established. Section IV analyzes the experimental results. Finally, Section V summarizes the research results and puts forward suggestions for future work.

II. Related Work

Automatic Train Operation (ATO) is a research hotspot in the field of railway transportation. Its core modules include train dynamics model, target speed curve optimization and speed tracking control. The essence of heavy haul train driving strategy optimization is to obtain a speed curve that meets the optimization goal. The following sections will introduce the research status of heavy haul train speed curve optimization and manipulation optimization using reinforcement learning.

A. Speed curve optimization of heavy haul train

Rao and Feng [4] designed the optimal feasible strategy based on the actual freight railway line analysis, and discussed the optimality and uniqueness of the analytical solution of train energy-saving operation. Bai and Xiao [5] proposed a multi-objective optimization model considering many practical constraints, which was solved by nonlinear approximation and quadratic programming (QP) algorithm, and the optimal trajectory was obtained. Li [6] proposed a time-limited multi-objective speed curve optimization model to achieve energy saving, punctuality and stability of train operation. Su [7] solved the problem of train trajectory optimization with STC by mixed integer quadratic programming (MIQP), and carried out actual case simulation research with real data. Dong [8] analyzed the driving difficulties of heavy haul trains in conventional sections and ramp sections, established a driving strategy constraint model, designed a train operation strategy generation method based on fuzzy predictive control, and locally optimized the train operation strategy of complex line sections to obtain the target operation curve. The above research on the optimization of the speed curve of heavy haul trains has not deeply explored or simplified the time-delay characteristics of air braking force. Especially in the scene of long and steep downhill, most of them are simulated by empirical formulas.

B. Train operation optimization based on reinforcement learning

Based on the Q-learning algorithm, Zhu [9] constructed a train eco-driving strategy for energy-saving goals, and introduced definitions related to travel time and energy consumption distribution in the state representation. The effectiveness and robustness of the proposed method are verified by deterministic and random environments. Due to the size limitation of the look-up table in the Q-learning algorithm, in the optimization problem with large state space, only the problem of ' dimension explosion ' can be solved by reducing the accuracy. Liu [10] proposed an intelligent control method for heavy haul trains running on long and steep downhill sections based on deep Q network (DQN) algorithm, and carried out experimental simulation with actual data to verify the effectiveness and robustness of the proposed DQN-based control method, but it also simplified the air brake. Shang [11] proposed a DRL method with a reference system (DRL-RS) for active handling of constraints, where the reference system is responsible for checking and correcting the learning progress of the agent to avoid going farther and farther on the wrong road. In order to achieve precise positioning and speed tracking control of the train position, Wang [12] proposed a method based on integral reinforcement learning (IRL) and parameter identification.

III. Problem Formulation

A. Train model

The multi-particle model of heavy haul train is established, and each vehicle is regarded as a particle.Vehicles are connected by coupler and draft gear, and the train only has longitudinal freedom by default. Assuming that there are $n+1$ vehicles in the train, the Equation 1 can be obtained by analyzing the longitudinal force of section i.

$$m_i\ddot{x}_i = F_{Ti} - F_{Bia} - F_{Bie} - F_{Ri} + F_{c(i-1)} - F_{ci} \quad (i = 0, 1, \ldots, n) \tag{1}$$

Among them, m_i is the quality of the i car, $\ddot{x}_i$ is its acceleration, F_{Ti} is the traction of the i car, only the locomotive can provide traction.F_{Bia} is the air braking force of the i carriage, F_{Bie} is the electric braking force of the i carriage, and only the locomotive is affected by the electric braking force.F_{Ri} is the resistance of the i carriage, F_{ci-1} represents the force between the adjacent previous carriage and the i carriage, F_{ci} is the force between the i carriage and the $i+1$ carriage.The air braking force will have an impact on the vehicle during the establishment and mitigation stages of air braking. In order to simulate the time-varying characteristics of air braking force in the establishment stage and the release stage, functions ${f_{\mathrm{delay1}}}^i(t)$ and ${f_{\mathrm{delay2}}}^i(t)$ [13] are introduced, and their expressions are shown in Equation 2 and Equation 3.

$$f^i_{\mathrm{delay1}}(t) = \begin{cases} 0, & t \le t^i_{\mathrm{delay1}} \\ \left(\dfrac{t^i_{\mathrm{delay1}} - t}{t_{\max 1}}\right)^{\lambda_1}, & t^i_{\mathrm{delay1}} \le t \le t_{\max 1} + t^i_{\mathrm{delay1}} \\ 1, & t_{\max 1} + t^i_{\mathrm{delay1}} < t \end{cases} \tag{2}$$

Authorized licensed use limited to: University of Pittsburgh. Downloaded on July 07,2026 at 23:51:09 UTC from IEEE Xplore. Restrictions apply.

$$f^i_{\text{delay2}}(t) = \begin{cases} 1, & t \le t^i_{\text{delay2}} \\ \left[1 - \left(\dfrac{t^i_{\text{delay2}} - t}{t_{\max 2}}\right)\right]^{\lambda_2}, & t^i_{\text{delay2}} \le t \le t_{\max 2} + t^i_{\text{delay2}} \\ 0, & t_{\max 2} + t^i_{\text{delay2}} < t \end{cases} \tag{3}$$

Among them, t^i_{delay1} is the delay time of the i vehicle responding to the air brake command, $t_{\max 1}$ is the time when the brake cylinder pressure rises from 0 to the maximum value, t^i_{delay2} is the delay time of the i vehicle responding to the relief command, and $t_{\max 2}$ is the time when the brake cylinder pressure drops from the maximum value to 0. The air braking force of the train at time t is obtained by multiplying the basic air braking force defined in Equation 4 and Equation 5.

$$B = \varphi_h \cdot \sum K_h \tag{4}$$

$$F_{Bia} = B \cdot f^i_{\text{delay}}(t) \tag{5}$$

φ_h is the braking shoe conversion friction coefficient, K_h is the train conversion braking shoe pressure. The electric braking force F_{Bie} can be obtained by the empirical formula Equation 6, where a_e is the electric braking gear.

$$F_{Bie} = \begin{cases} 153.3 \cdot v \cdot a_e, & 0 \le v \le 3 \\ 461 \cdot a_e, & 3 < v \le 75 \\ 34560 \cdot a_e / v, & 75 < v \end{cases} \tag{6}$$

In this paper, the traction force F_{Ti} is also obtained by the empirical formulaEquation 7 :

$$F_{Ti} = \begin{cases} 760, & 0 \le v \le 5 \\ 779 - 3.8 \cdot v, & 3 < v \le 75 \\ 34560/v, & 75 < v \end{cases} \tag{7}$$

B. Optimization algorithm based on Dueling DQN

Dueling DQN changes the single-branch network of DQN into a double-branch structure [14].As shown in Figure 1, the Q-value function is decomposed into a state value function V and a dominance function A Equation 8 to estimate the value of all actions in a certain state and the advantage of an action relative to other actions. This can better evaluate the value of the state and speed up the training. The driving operation process of heavy haul train involves a large number of state space and action space, which is suitable for optimization with Dueling DQN algorithm.

$$Q(s,a;\theta,\alpha,\beta) = V(s;\theta,\beta) + \left(A(s,a;\theta,\alpha) - \frac{1}{|A|}\sum_{a'} A(s,a';\theta,\alpha)\right) \tag{8}$$

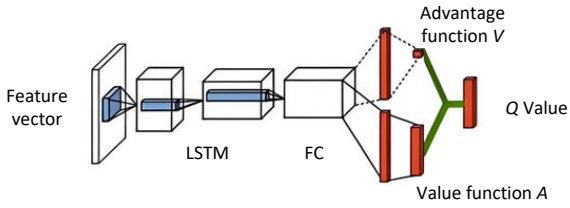


Fig. 1: The network structure of Dueling DQN

1) States and action spaces: In this study, the state refers to the current running information of the train, including : running time T, displacement $X_{k,n\times n}$, speed $V_{k,n\times n}$ and acceleration $a_{k,n\times n}$ of each vehicle. In addition, this study uses the empty walking stage, transient stage and steady state stage to simulate the establishment and release process of air braking. Therefore, the information representing the state of air braking needs to be added to the state space : single air braking duration t_a, whether the previous action is air braking f_a, whether it is currently alleviating f_r, and relieving countdown t_r. Then the state space can be expressed as Equation 9.

$$S_k = \{T,\ X_{k,n\times n},\ V_{k,n\times n},\ a_{k,n\times n},\ t_r,\ t_a,\ f_r,\ f_a\} \tag{9}$$

The actions that Agent can take include : traction,electric braking and air braking, where traction and air braking are represented by two Boolean variables a_t and a_a, respectively. The electric braking a_e adopts the method of grading, a total of 6 gears, and the action space can be expressed as Equation 10 .In this paper, it is assumed that the train only adopts the common air braking with 50 kPa decompression.

$$A_k = \{a, a_e, a_a\} \quad (a_t \wedge a_e = 0,\ a_t \wedge a_a = 0,\ a_e \in [0, 0.2, 0.4, 0.6, 0.8, 1]) \tag{10}$$

2) Constraints:

- Safety : it is the most important indicator in the operation of heavy haul trains. The speed of the train during operation must be within the specified speed range, such as Equation 11.

$$V_{\min} \le V \le V_{\max} \tag{11}$$

- The minimum air filling time : the heavy haul train will adopt the circulating air brake to control the train not to exceed the speed in the long downhill.When a certain air braking of the train ends, it enters the release stage. At the same time, the air cylinder of the train begins to fill the air. After the pressure in the air cylinder meets the requirements, the next air brake can be implemented. Otherwise, it will not be able to provide enough air braking force to the train, resulting in overspeed of the train. For 20,000 tons of heavy haul trains, the minimum air filling time is 180 s [10]. Assuming that the train

Authorized licensed use limited to: University of Pittsburgh. Downloaded on July 07,2026 at 23:51:09 UTC from IEEE Xplore. Restrictions apply.

stops the j air brake at $T_{air,j}$, and the minimum air filling time is T_{AL}, then there is Equation 12.

$$T_{\text{air},j+1} - T_{\text{air},j} \geq T_{\text{AL}} \tag{12}$$

3) Evaluation indicators:

- Average speed : In this paper, the average running speed of the train in the set time is used to measure the efficiency of the train operation. The higher the average speed, the higher the efficiency of the train operation. Assuming that the total running time is T, and the acceleration of the train at time t is a, the average speed V_{avg} of the train during operation is defined as Equation 13 :

$$V_{\text{avg}} = \frac{1}{T}\int_0^T a(t)\,dt \tag{13}$$

- The total time of air braking : In this study, the time of air braking during the operation of the train is taken as an indicator to measure how much air braking is taken by the train. The longer the total time, the longer the friction time of the train brake shoe, and the greater the loss of the train Equation 14 :

$$T_s = \int_0^T a_a dt \tag{14}$$

4) Reward function and termination conditions:

- The algorithm rewards include : safety rewards, speed rewards, air brake penalties, such as type Equation 15. Among them, r_v is the speed reward, V is the current average speed of the train, when the train is speeding, give a negative reward C_1, otherwise give a positive reward D ; r_a is the air brake reward. When the train adopts air brake, it is given a negative reward C_2, otherwise the reward is 0.

$$\begin{aligned} R_k &= r_v + r_a \\ r_v &= \begin{cases} C_1, & V < V_{\min} \,\|\, V > V_{\max} \\ D, & V_{\min} < V < V_{\max} \end{cases} \\ r_a &= \begin{cases} C_2, & a_a = 1 \\ 0, & a_a = 0 \end{cases} \end{aligned} \tag{15}$$

- The termination condition is very important for whether the optimization algorithm can converge with the training. In this study, if the train speed exceeds the safe range, this situation is unacceptable. At this time, the termination condition Done is set to True, and the other termination condition is that the train running time reaches the set upper limit.

The specific process of DuelingDQN is shown in algorithm1, and the overall framework is shown in Figure 2.

IV. Experimental Results

A. Experimental Environment

The heavy haul train formation mode in this paper is ' 1HXD1 + 108C80 + 1HXD1 + 108C80 ', the total length

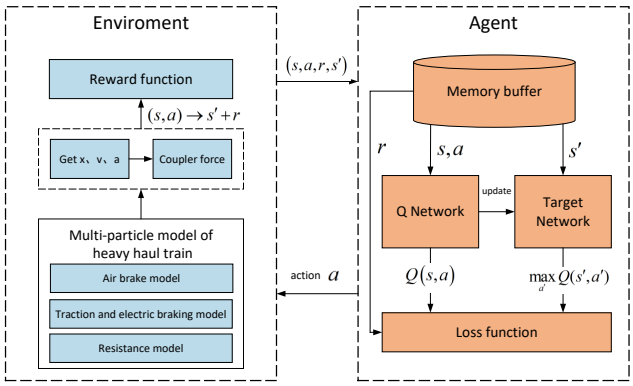


Fig. 2: Optimization algorithm structure diagram

Algorithm 1 Dueling DQN Training Algorithm

Step 1: Initialization
Random weight initialization for Dueling DQN evaluation network
Initialize target network with same structure and weights
Initialize experience pool (size S), discount rate γ, exploration rate ϵ, exploration decay rate$g$, learning rate λ, batch size n_e, target update interval K
Calculate total steps based on planned running time and time interval

Step 2: Training
for episode $= 1$ **to** M **do**
  Initialize performance parameters, train status, environment parameters, line speed limit
  for step $= 1$ **to** N **do**
    Select action a using ϵ-greedy policy
    Execute a, state transferred from s to s', reward r, and done flag
    Store transition $(s, a, r, s', \text{done})$ in experience pool
    Sample minibatch of size n_e from experience pool
    if done $=$ True **then**
      $y = r$
    else
      $y = r + \gamma \max_{a'} Q(s', a'; \theta^-)$
    end if
    Compute gradient with respect to weights θ
    Update evaluation network parameters using SGD
    if done **then**
      break
    end if
    if episode $\equiv 0 \pmod K$ **then**
      Update target network weights: $\theta^- \leftarrow \theta$
    end if
  end for
end for
Save the trained Dueling DQN model

Authorized licensed use limited to: University of Pittsburgh. Downloaded on July 07,2026 at 23:51:09 UTC from IEEE Xplore. Restrictions apply.

of the line is 15000m, the slope of the line is -10.2 ‰, and the model training is carried out on a desktop PC with a processor speed of 3.4GHz and a memory size of 16GB.

B. Parameters Setting

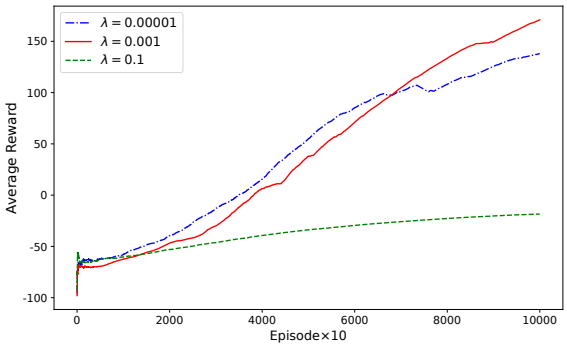
For RL, selecting the appropriate hyperparameters can improve training efficiency and prevent instability during training.This paper explores the influence of learning rate λ and exploration attenuation rate g on the training effect. From Figure 3a , it can be seen that when training 100,000 rounds, λ is 0.001 and 0.00001, the cumulative reward is increasing. After 70,000 rounds, the model reward of λ 0.001 increases faster, but when λ is 0.1, the reward increases slowly and gradually stabilizes at about -18, which indicates that the model may fall into a local optimal situation. Figure 3b shows different effects of g. When g is 0.9, the exploration rate decays too fast, and the model relies too early on the estimated value of the current Q value, resulting in the agent failing to fully explore all possibilities of the state space and falling into the local optimal dilemma. Although the model with g of 0.99995 increased slowly in the early stage, it explored the state space more fully, and the reward gradually increased and showed a convergence trend in the later stage of training. Therefore, we choose λ as 0.001, g as 0.99995.In order to improve the training efficiency, ΔT is set to 10 s, and other parameters are set as shown in Table I.

TABLE I: Parameters setting

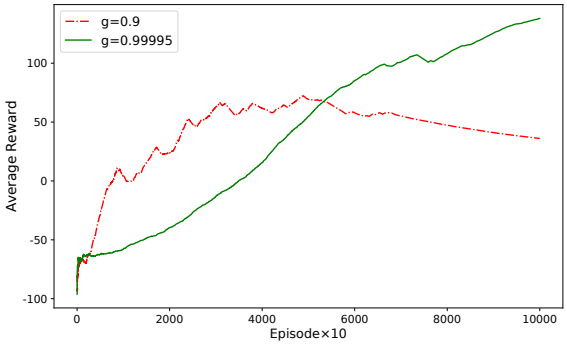
Parameter	Value
Maximum training episode M	2.5×10^5
Running time of one episode T	700 s
Update period of target network K	1000
Discount rate γ	0.95
Learning rate λ	0.001
Initial value of ε_0	1.0
Explore the decay rate g	0.999 95
Final value of $\varepsilon_{\min}$	0.02
Capacity of memory buffer S	50 000
Batch size of sampling n_e	256
Negative reward C_1	-100
Negative reward C_2	-2
Positive reward D	$\nu/10$
Minimum air-refilling time T_{AI}	180 s

C. Optimization Results

Figure 4 shows the change of the average cumulative reward after training 400,000 rounds. The blue curve represents the Dueling DQN algorithm and the red curve represents the DQN algorithm. We can see that the reward of Dueling DQN algorithm has converged and stabilized at about 188 at the time of about 310000 rounds. At this time, the reward value of DQN algorithm is 168, and in the subsequent training process, the reward of DQN algorithm is still increasing, indicating that DQN algorithm has not reached convergence in the end. On the whole, the increase rate of the average cumulative reward of the Dueling DQN algorithm is also



(a) Different λ



(b) Different g

Fig. 3: The average total reward curve under different hyperparameters

higher than that of the DQN algorithm, which indicates that the Dueling DQN algorithm is a better choice.

The blue curve in Figure 5 is the train v-t curve optimized by Dueling DQN algorithm,and the vehicle speed is based on the head car.. It can be seen that the time for the train to apply three times of air braking is 110 s, 370 s and 620 s, respectively. Because the air braking force has the characteristics of time delay, the train is only affected by the air braking force after the delay of 10 s, and the train speed begins to decline. The end times of the three air brakes are 190 s, 440 s and 650 s, respectively. The time between the two air brakes satisfies the constraint of the minimum air filling time Equation 12. After the output of air braking force in the remission stage is 0, the train speed begins to increase again. During the whole operation, the maximum train speed is 74.94 km/h, and the minimum is 51.39 km/h, which is within the safe range Equation 10. The green curve in Figure 5 is the train v-t curve optimized by the DQN algorithm. The train also performed three cycles of air braking. However, due to the different time for the train to reach the mitigation speed under the two algorithms, the speed will alternately rise.In this paper, Table II is used to compare the optimization effects of the two algorithms more intuitively. It can be seen that compared with the DQN

Authorized licensed use limited to: University of Pittsburgh. Downloaded on July 07,2026 at 23:51:09 UTC from IEEE Xplore. Restrictions apply.

algorithm, when the Dueling DQN algorithm is used to optimize, under the premise of ensuring safety, the average running speed of the train is higher and the total time of air braking is less. Therefore, the optimization effect of the Dueling DQN algorithm is better.

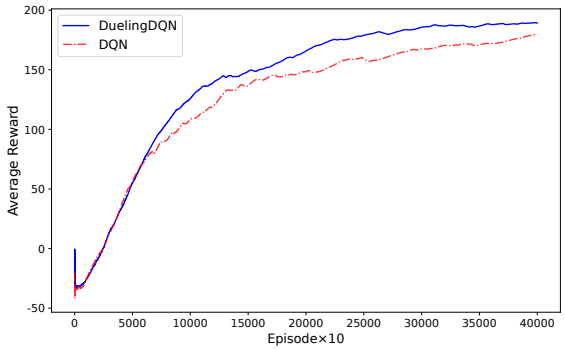


Fig. 4: The average reward curve of Dueling DQN and DQN

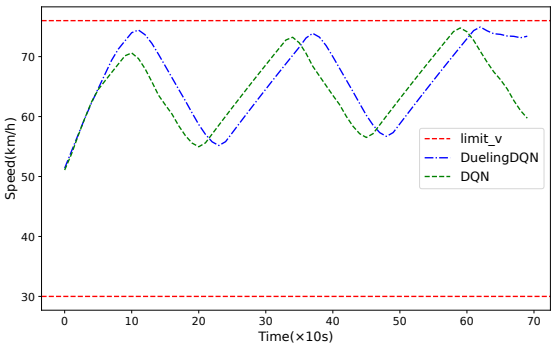


Fig. 5: Train v-t diagram under Dueling DQN and DQN

TABLE II: Comparison of optimization effects under different v_0

v_0 (m/s)	Approach	Evaluation indicators	
		V_{avg} (m/s)	T_s (s)
13.89	Dueling DQN	18.32	160
	DQN	17.63	180
16.67	Dueling DQN	19.88	170
	DQN	19.38	200

V. Conclusion

In this paper, a driving strategy optimization method for heavy haul trains based on Dueling DQN algorithm is proposed. Considering the time-delay characteristics of air braking, a multi-particle model of heavy haul trains is established. Based on the optimization goal of improving train operation efficiency and reducing the total time of air braking, the reward function is set. The simulation is carried out on the long downhill, and the effectiveness of the proposed algorithm is verified by comparison with the DQN algorithm. In order to improve the efficiency of model training, the following two measures are taken : the choice of time granularity is large, and 13 trailers are regarded as one section for processing, which leads to the reduction of simulation accuracy. These two problems need to be solved in the future work, and the actual train operation strategy should be further considered.

References

[1] A. I. Istamar, Terence, S. Philip, and Hidayaturrahman, "An autonomous player agent for game balance insight on an educational video game," in *2023 4th International Conference on Artificial Intelligence and Data Sciences (AiDAS)*, 2023, pp. 269–273.

[2] T. K. Saini, A. Maurya, D. Dhyani, and A. Saini, "Intricacy of cnn and rl with substantiation of cnn model in simulated self-driving environment," in *2025 International Conference on Cognitive Computing in Engineering, Communications, Sciences and Biomedical Health Informatics (IC3ECSBHI)*, 2025, pp. 866–870.

[3] H. Nguyen and H. La, "Review of deep reinforcement learning for robot manipulation," in *2019 Third IEEE International Conference on Robotic Computing (IRC)*, 2019, pp. 590–595.

[4] Y. Rao, X. Feng, Q. Wang, and P. S. and, "The optimal solution to the energy-efficient train control in a multi-trains system-part 1: the algorithm design," *Transportmetrica A: Transport Science*, vol. 21, no. 2, pp. 1–29, 2025.

[5] B. Bai, Z. Xiao, Q. Wang, P. Sun, and X. Feng, "Multi-objective trajectory optimization for freight trains based on quadratic programming," *Transportation Research Record*, vol. 2674, no. 11, pp. 466–477, 2020.

[6] W. Li, S. Jiang, and M. Jin, "Multiobjective optimization and weight selection method for heavy haul trains trajectory," *IEEE Access*, vol. 10, pp. 41 152–41 163, 2022.

[7] S. Mingliang, N. Xingliang, Z. Zipei, S. Pengfei, W. Qingyuan, and B. Baoxue, "Trajectory optimization for freight train with section time constraints using mixed integer quadratic programming," in *2022 IEEE 17th Conference on Industrial Electronics and Applications (ICIEA)*, 2022, pp. 1545–1551.

[8] J. Dong, H. Yu, G. Tai, F. Li, and Y. Huang, "Research on the driving strategy of heavy-haul train based on fuzzy predictive control," in *2020 IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC)*, 2020, pp. 1–7.

[9] Q. Zhu, S. Su, T. Tang, W. Liu, Z. Zhang, and Q. Tian, "An eco-driving algorithm for trains through distributing energy: A q-learning approach," *ISA Transactions*, vol. 122, pp. 24–37, 2022.

[10] W. Liu, S. Su, T. Tang, and X. Wang, "A dqn-based intelligent control method for heavy haul trains on long steep downhill section," *Transportation Research Part C: Emerging Technologies*, vol. 129, p. 103249, 2021.

[11] M. Shang, Y. Zhou, and H. Fujita, "Deep reinforcement learning with reference system to handle constraints for energy-efficient train control," *Information Sciences*, vol. 570, pp. 708–721, 2021.

[12] J. Wang, S. Gao, H. Dong, and Z. Zhang, "Adaptive control for automatic high-speed trains operation by integral reinforcement learning and parameter identification," in *2019 Chinese Control Conference (CCC)*, 2019, pp. 6645–6650.

[13] W. Liu, S. Su, T. Tang, and Y. C. and, "Study on longitudinal dynamics of heavy haul trains running on long and steep downhills," *Vehicle System Dynamics*, vol. 60, no. 12, pp. 4079–4097, 2022.

[14] Z. Wang, T. Schaul, M. Hessel, H. Hasselt, M. Lanctot, and N. Freitas, "Dueling network architectures for deep reinforcement learning," in *Proceedings of The 33rd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, M. F. Balcan and K. Q. Weinberger, Eds., vol. 48. New York, New York, USA: PMLR, 20–22 Jun 2016, pp. 1995–2003.

Authorized licensed use limited to: University of Pittsburgh. Downloaded on July 07,2026 at 23:51:09 UTC from IEEE Xplore. Restrictions apply.